K BALAJI

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PROFESSIONAL SUMMARY

Computer Science Engineering graduate with internship experience in software development, Python automation, API testing, Android development, databases, and cloud-readiness initiatives. Proficient in Python, SQL, Java, data analysis, machine learning, REST APIs, backend development, and AI/LLM technologies, with hands-on experience building secure payment, IoT, healthcare analytics, and mobile applications. Strong foundation in software engineering, data structures, database management, cloud computing, API integration, and end-to-end system development.

TECHNICAL SKILLS

Languages: Python, Java, Kotlin, JavaScript, HTML5, CSS3.
Data & Analytics: Pandas, NumPy, Matplotlib, Seaborn, Microsoft Excel, Power BI, Tableau.
AI/ML/LLM Tools: LangChain, RAG, LlamaIndex, OpenAI API, Hugging Face Transformers, Ollama, n8n.
Backend & Web: Django, Flask, RESTful APIs, React, Node.js, Express.js.
Databases: MySQL, MongoDB, Firebase Firestore.
Cloud, DevOps & Tools: Microsoft Azure, Git, GitHub, Linux.
Mobile & IoT: Android studio, Jetpack Compose, Arduino, Wireless Communication.

PROFESSIONAL EXPERIENCE

Android App Development Intern — MindMatrix.io *Dec 2025 – May 2026*
Bangalore, Karnataka

- Developed an Android folk artist discovery and booking platform using Kotlin, Jetpack Compose, Firebase Firestore,and repository architecture with CRUD operations, search filters, offline-friendly handling, and real-time discovery.
- Implemented troupe profiles, portfolio galleries, district/art-type filtering, and one-tap Call/WhatsApp booking workflows to connect folk artists with event planners.

Software Developer Intern — Mindcloudz Innovation *Jan 2025 – June 2025*
Bangalore, Karnataka

- Developed Python automation scripts to streamline service-request workflows, reducing manual workload by 20% and improving process efficiency.
- Created database-population and API-testing tools to support cloud readiness and software-component migration within the Amadeus backend environment.

PROJECTS

Biometric-Enabled Merchant Payment System
Python, SQL, Raspberry Pi, REST APIs, Biometric Authentication, IoT

- Engineered a secure biometric POS payment system using Raspberry Pi and biometric sensors, strengthening merchant authentication and reducing reliance on conventional PIN-based verification.
- Developed Python backend services with REST API integration and SQL database management for transaction processing, authentication workflows, real-time records, validation, error handling, and security.

IoT-Based Real-Time LED Matrix Display
Arduino, C/C++, IoT, Wireless Communication, API Integration

- Developed an IoT-enabled LED matrix system that retrieves and displays real-time internet-based information through automated wireless data updates.
- Programmed Arduino firmware for data processing, LED matrix control, dynamic content rendering, device operations, debugging, hardware integration, and end-to-end testing.

COVID-19 Hospital Bed Management & Analytics System
MySQL, SQL, Python, DBMS, Data Analysis, Data Visualization

- Designed a centralized healthcare database for tracking hospital beds, admissions, discharges, occupancy, and availability across multiple facilities to support resource allocation.
- Designed normalized schemas and optimized SQL queries using joins, aggregations, subqueries, views, constraints, and indexing; analyzed occupancy trends using Python and visualization techniques.

EDUCATION

Bachelor of Engineering — Computer Science Engineering *2022 – 2026*
ACS College of Engineering, Bengaluru **CGPA: 8.1**

CERTIFICATIONS & ACHIEVEMENTS

- **Oracle Certified Foundations Associate** — Oracle University (2025)
- **AI-900 & AI-102 Microsoft Certified: Azure AI Fundamentals**
- **EY and Microsoft — AI Skills Passport Certification**
- **Green Skills and Artificial Intelligence** — Edunet Foundation Skills4Future Program
- **MERN Stack Web Application** — Acadoemor
- **Research Paper Published** — Biometric Payment System through Microsoft CMT, focusing on secure and seamless authentication for digital transactions